



State-of-health (SOH) evaluation on lithium-ion battery by simulating the voltage relaxation curves

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ABSTRACT

State-of-health evaluation of lithium ion batteries is one of the key challenges to construct efficient battery system with enhanced performance. In this paper, we demonstrate a facile evaluation method which is based on the voltage stabilization behaviour during relaxation. A second-order equivalent circuit is employed to describe the voltage relaxation. By simulating the voltage relaxation curves, the characteristic parameters that can distinguish the ohmic, activation and concentration depolarizations are acquired. The obtained parameters are sensitive to electrochemical states, which provide comprehensive evaluation to the state-of-health of batteries and aging mechanism analysis. As an example, the proposed method identifies two aging stages of $\text{Li}(\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3})\text{O}_2/\text{graphite}$ pouch cells during accelerated aging. Namely, a loss of lithium inventory in the first stage of aging, and the loss of active material in the next stage. Postmodern analysis including X-ray diffraction and high-resolution transmission electron microscope well agrees with the voltage relaxation simulation. The proposed state-of-health evaluation method is non-destructive and accessible for on-board aging diagnosis, which is of great significance for developing better battery management system.

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1. Introduction

The state-of-health (SOH) evaluation of lithium ion batteries (LIBs) is one of the urgent tasks for battery systems, especially in the large scale applications such as electrical vehicles and grid storage system [1]. Accompanying with aging, LIBs experience capacity loss, resistance increase and voltage decay, which makes battery management difficult during subsequent charging and use. Accurate diagnosis of battery status will help to modify charging strategies to avoid abuse or even thermal runaway. In this regard, the key parameters for SOH evaluation are of great importance. Generally, excellent SOH parameters require three features: firstly, the parameters should describe electrochemical states accurately.

Then, they should be easily obtainable (online prefer) and non-destructively acquired. Thirdly, they can be used to analyse the possible aging factors and mechanisms to help develop preventive solutions [2–6]. So far, capacity and resistance are the mostly used parameters describing the SOH of LIBs. Many researchers have made elaborate efforts on actual capacity evaluation, resistance identification (DCR, ASI) and their evolutionary behaviour during cycling or storage in different types of LIBs [7–10]. Although those parameters can directly indicate the battery performance, people cannot get more information in relation to the inner degradation mechanisms, which restricts our analysis and understanding on aging. Other parameters also reported in literature, which are listed in Table 1.

For example, electrochemical impedance spectroscopy (EIS) is an effective analytical method for identifying the internal processes of LIBs. It provides abundant information on the electrode reactions [11,12]. Researchers can distinguish bulk resistance (R_b), charge-transfer resistance (R_{ct}) and Warburg impedance (Z_w) through suitable circuit model to analyse internal decay mechanisms with respect to the solid electrolyte interface (SEI) and electrochemical

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Table 1
Parameters and methods used in references for SOH evaluation.

Parameters/Methods	Feature
Capacity-based	Easy to obtain, simple evaluation
DC resistance (DCR)	Easy to obtain, simple evaluation
Area-specific impedance (ASI)	Easy to obtain, simple evaluation
AC impedance (EIS)	Hard to obtain, In-depth evaluation, expensive
Incremental capacity (dQ/dV)	Hard to obtain, In-depth evaluation, off-line
Differential voltage (dQ/dV)	Hard to obtain, In-depth evaluation, off-line

performance [13–18]. Unfortunately, EIS tests rely on complicated and expensive equipment which make it hard to apply in a real battery module. Another shortcoming is that the ultra-low resistance of EVs or PHEVs batteries are almost beyond the detection range of EIS and therefore reliable results are difficult to acquire. Some methods based on *in-situ* voltage measurement have been proposed, such as incremental capacity (dQ/dV) and differential voltage (dV/dQ) [16,19–27]. By charging or discharging the battery at low constant current (for example, C/25 rate), the relationship between capacity and cell potential under quasi-equilibrium condition are acquired. Subtle changes in the charge/discharge curve can be accentuated and easily analysed after mathematical processing, such as differentiation. By this way, the main aging factors, such as loss of lithium inventory (LLI) or loss of active materials (LAM) [27,28], can be identified and assessed. However, how to achieve online and rapid evaluation still needs further exploration.

To date, novel methods and parameters for SOH evaluation are still in pressing demands. LIBs are subjected to three states during their entire life: charge, discharge and rest. The voltage relaxation (VR) behaviours during rest, an important characteristic which could be easily obtained by *in-situ* voltage measurement, is rarely noticed. In this work, we develop a SOH evaluation method based on VR curve simulation. Through modelling the VR behaviour using a second-order RC electrical circuit, the important parameters including stable voltage (V_s) and three kinds of resistances (R_o , R_a , and R_c) in relation to the ohmic depolarization, the activation depolarization and the concentration depolarization processes, respectively, are identified and used to assess the SOH of LIBs. We applied this method on NCM coin type cells as well as commercial

NCM/graphite LIBs during accelerated aging test. The validity and practicability of the proposed method are further compared with non-destructive electrochemical analysis, including DC internal resistance test (DCR), Differential voltage (dQ/dV) analysis, and postmortem analysis including specific capacity examination, X-ray diffraction and high-resolution TEM. It is suggested that this novel method is a promising candidate for non-destructive and online aging diagnosis and SOH evaluation.

2. The origin of VR curve and key parameters acquisition

As is known to all, charging and discharging will cause the disequilibrium in the distribution of electrons and ions in the electrodes, which is called polarization. The polarization will induce the electrode potential deviated from equilibrium potential. Overall, there are three main forms of polarization giving rise to the voltage deviation [29,30]. One is the ohmic polarization related to electrons which causes the voltage drop since the physical resistance of the electrode, the electrolyte and the separator. Second is the activation polarization associated with the Li-ions in the charge-transfer process. Last is concentration polarization depending on the Li-ions in solid phase diffusion. When the charging/discharging processes stopped, the voltage of LIBs still varies during relaxation as shown in Fig. 1a. This can be understood as a charge redistribution process in the electrodes or the so-called depolarization process. According to the understanding of polarization phenomena in the charging/discharging processes, the depolarization process can be divided into three processes: the ohmic depolarization, the activation depolarization and the concentration depolarization. The activation depolarization and concentration depolarization are related to the charge balance at particle-electrolyte interface and in the particle bulk, respectively (Fig. 1b). In this regard, voltage relaxation behaviour contains direct information with respect to the lithium ions and electrons transportation. It can be taken as an indicator on SOH evaluation. Fig. 1c display a typical VR curve for a Li-ion battery which was discharged to 3.3 V at 1C-rate and then rest. At the end of the discharge, there is a drastic change of cell voltage from cut-off voltage V_0 to V_1 , which is due to the quick response and redistribution of electrons (i.e. the ohmic depolarization). From V_1 to V_2 , the cell voltage rises quickly

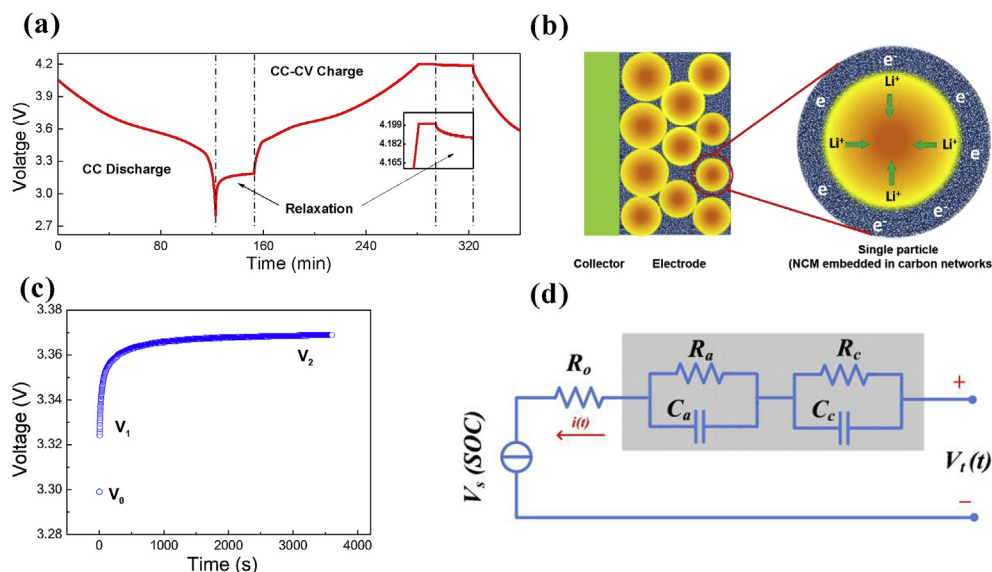


Fig. 1. (a) Voltage varies with time during charge, discharge and relaxation in NCM/C Li-ion battery; (b) Single particle model for NCM cathodes during discharge process; (c) A typical voltage relaxation curve during rest; (d) The second order RC model which is applied to describe V-t characteristics of VR behaviour.

at initial state and then slowly reaches a steady value, which is caused by the activation and the concentration depolarizations. Due to the kinematic difference of Li-ions diffusion at the electrode-electrolyte interface and in the bulk of active materials, the VR process is dominated by one of the depolarizations with relaxation time prolonging. Time constant τ (in seconds) is used to characterize the time response of different steps. The relationship of the three time constants is: $\tau_o < \tau_a < \tau_c$ (correspondence to the ohmic depolarization, the activation depolarization and the concentration depolarization, respectively).

In order to describe the VR curves quantitatively, an equivalent circuit model is adopted to describe the V - t characteristic during rest. Fig. 1d displays the second-order RC model. V_s is the theoretical voltage in steady state, which strongly depends on the State-of-Charge (SOC) at a certain temperature. A series resistor R_o is used to describe the instantaneous voltage change related to the ohmic depolarization. The RC parallel network in shade is used to predict the transient response. Specifically, the parallel network of R_a and C_a is responsible for the short time constant (τ_a) process caused by activation depolarization, while the parallel network of R_c and C_c is responsible for the long time constant (τ_c) process caused by concentration depolarization.

Therefore, the battery voltage $V_t(t)$ can be calculated as equation (1). $U_a(t)$ and $U_c(t)$ represent the voltage variation with time caused by activation depolarization and concentration depolarization, respectively. R_o represents the ohmic depolarization resistance. Suppose I is the charge or discharge current. When $t = 0$, $i(0) = I$; when $t > 0$, $i(t) = 0$.

$$V_t(t) = V_s - i(t)R_o - U_a(t) - U_c(t) \quad (1)$$

During the relaxation, the differential form of voltage vs. time is shown in (2) and (3). When $t = 0$, $U_a(0) = V_a = IR_a$, $U_c(0) = V_c = IR_c$.

$$\frac{dU_a}{dt} = -\frac{U_a}{R_a C_a} + \frac{I}{C_a} \quad (2)$$

$$\frac{dU_c}{dt} = -\frac{U_c}{R_c C_c} + \frac{I}{C_c} \quad (3)$$

In the condition that the batteries first discharge with a constant current (i.e. $I > 0$) and then rests, $U_a(t)$ and $U_c(t)$ is expressed by (4) and (5).

$$U_a(t) = V_a e^{-\frac{t}{\tau_a}} \quad (4)$$

$$U_c(t) = V_c e^{-\frac{t}{\tau_c}} \quad (5)$$

where $\tau_a = R_a C_a$ and $\tau_c = R_c C_c$ represent the time constants of activation polarization and concentration depolarization, respectively.

Based on equations (1)–(5), the VR model is described as equation (6):

$$V_t(t) = V_s - i(t)R_o - V_a e^{-\frac{t}{\tau_a}} - V_c e^{-\frac{t}{\tau_c}} \quad (6)$$

$$V_t(t) = V_s - V_a e^{-\frac{t}{\tau_a}} - V_c e^{-\frac{t}{\tau_c}} \quad (7)$$

($V_a = IR_a$, $V_c = IR_c$, the value of I is positive when discharge).

Since the ohmic depolarization terminates instantaneously at the end of charging/discharging, the actually recorded VR data mainly reflect the process of activation depolarization and concentration depolarization. So the fitting formula for actual data should be written as equation (7). According to the second-order RC model, the VR curves can be simulated by nonlinear fitting with

Table 2

Parameters obtained from VR curve fitting and their electrochemical meaning in SOH analysis.

Parameters	Computation	Description
R_{total}	$R_{total} = V_s - V_t(0) /I$	The total resistance of the battery
R_o	$R_o = R_{total} - \frac{R_a - R_c}{I}$	Resistance in relation to ohmic depolarization
R_a	$R_a = V_a/I$	Resistance in relation to activation depolarization
R_c	$R_c = V_c/I$	Resistance in relation to concentration depolarization
τ_a	/	Time constant of activation depolarization
τ_c	/	Time constant of concentration depolarization
V_s	/	The voltage of the cell when it reaches a steady state during relaxation

equation (7) to obtain the values of V_s , V_o , V_a , V_c , τ_a and τ_c . Other parameters on resistances, such as R_o , R_a , R_c and R_{total} , can also be accurately calculated to evaluate the SOH of LIBs. Those parameters used to evaluate SOH and the calculation method are listed in Table 2. In supporting information, Fig S1 and S2 compared the VR data fitting results based on second-order RC model and traditional Thevenin-based model. The goodness-of-fit (R^2) is 0.9951 with proposed second-order RC model, which is better than Thevenin-based model (0.9316). This result suggests that the proposed fitting method provides high precision results. And also, the effectiveness of the model parameters was proved by the comparison of cyclic voltammetry (CV) test and VR fitting at different state-of-charge, which can be seen in the supporting information (Fig. S6 and Table S1).

3. Capacity estimation of NCM electrode by VR simulation

The VR simulation method was firstly investigated on the capacity estimation of NCM electrodes in half-cells. The NCM electrodes is composed of 80 wt % $\text{LiNi}_{0.33}\text{Co}_{0.33}\text{Mn}_{0.33}\text{O}_2$, 10 wt% PVDF (polyvinylidene fluoride) and 10 wt % carbon black with a load of 4.5 mg cm^{-2} . The electrolyte is 1M LiPF_6 in EC (Ethylene carbonate), DMC (Dimethyl carbonate) and EMC (Ethyl methyl carbonate) solution (1:1:1 in volume). The half cells were cycled for three cycles at a rate of 0.2 C and then cycled at 1 C rate under 25°C . A 10 min rest was performed at the end of each charging and discharging. Fig. 2a displays the cycling performance. The capacity of 140 mAh g^{-1} was delivered at the first several cycles at 1C rate. After 500 cycles, 100 mAh g^{-1} is still maintained, which shows very stable cycling property. Fig. 2b and c displays the evolution of VR curve with different cycles. At the end of charge state, the voltage decrease with rest in dozens of mV (Fig. 2b), while at the end of discharge state, the voltage increase in hundreds of mV (Fig. 2c). This is due to the different polarization and depolarization in charge and discharge states, which is discussed in supporting information (Fig. S3). With the cycling and aging of the cells, the polarization and depolarization increase and affect on both charged and discharged states. Therefore, the gradual degradation of VR curve can be seen after 50, 150, 250, 350 and 450 cycles in Fig. 2b and c. Through the VR simulation, this behaviour can be investigated quantitatively. Table 3 list the key parameter of stable voltage (V_s) and resistances. It is interesting that the evolution of actual capacity versus stable voltage displays a very good linear relationship. And it is stable at both charge and discharge states, which indicated the possibility of capacity estimation by VR simulation. It is worth noting that, as shown in Fig. 2c, large depolarization results in a transient change that will reduce the fitting accuracy of 2.7 V. Therefore, it is recommended to select the appropriate SOC for data acquisition, fitting and comparison of the SOH states.

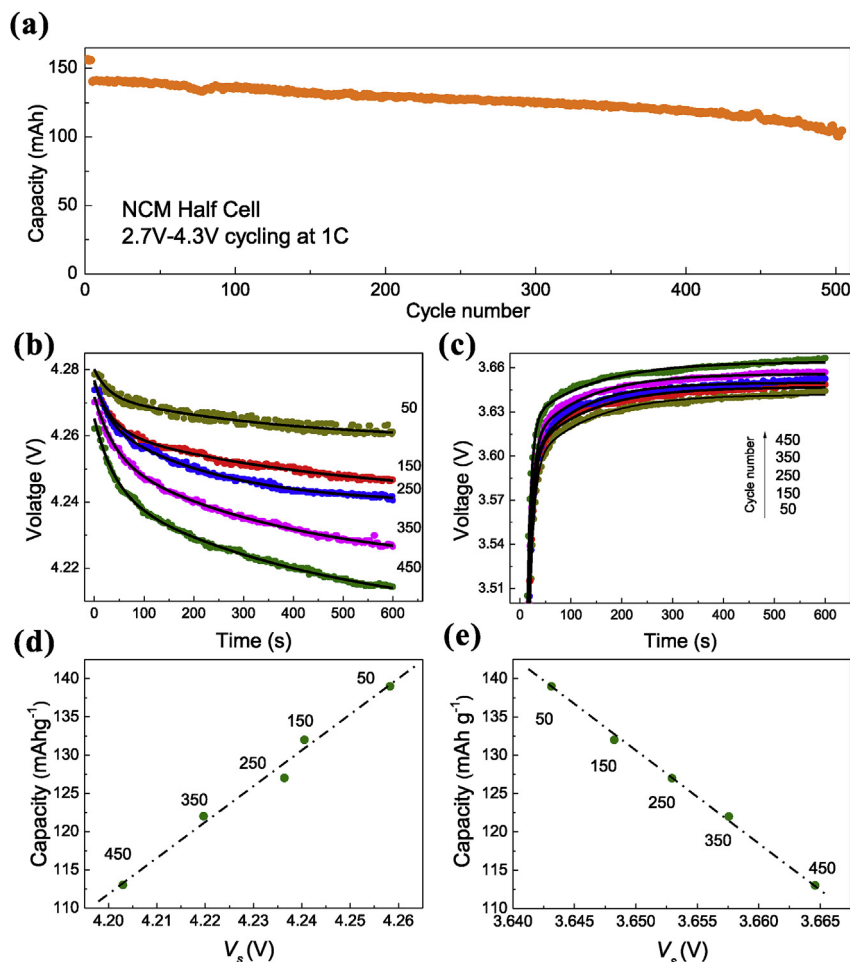


Fig. 2. (a) Cycling performance of NCM half-cell at 1C rate; Simulation of VR curves at 50, 150, 250, 350 and 450 cycles: (b) after charging to 4.3 V and (c) after discharging to 2.7 V; (d) and (e) The linear relationship between stable voltage and capacity of NCM material.

Table 3

The key parameters obtained by VR simulation in Fig. 2b and c.

Cycle number		50	150	250	350	450
4.3 V	R^2	0.9341	0.9918	0.9951	0.9973	0.9975
	V_s	4.258	4.241	4.239	4.220	4.203
	V_o	28.942	34.106	33.931	41.211	50.795
	V_a	9.292	19.737	16.937	25.631	31.685
	V_c	21.342	29.497	37.549	50.357	59.052
	V_{total}	59.576	83.340	88.417	117.199	141.532
2.7 V	R^2	0.9902	0.9893	0.9904	0.9894	0.9900
	V_s	3.643	3.648	3.653	3.658	3.665
	V_o	145.237	167.440	141.007	226.521	72.896
	V_a	1164.858	1152.604	1187.411	1111.568	1273.873
	V_c	65.748	63.252	62.4508	58.8038	60.365
	V_{total}	1375.842	1383.297	1390.868	1396.893	1407.133

4. Application of VR simulation on commercial LIBs

In order to verify the practical application of VR simulation method, an accelerated aging experiment was conducted using commercial NCM/graphite pouch cell. Under high temperature accelerated aging conditions, the SOH of the battery changes rapidly, and it can be easily found whether the VR method is sensitive to SOH during aging. The specifications of the full cells are listed in Table S2. In this experiment, different SOH evaluation

methods such as standard capacity test, DCR test, incremental capacity analysis and VR simulation, were carried out as well as the postmodern analysis. For the accelerated aging test, the full cells were cycled at 75 °C and 1C rate. After pre-set charge-discharge cycles, the cells were move to 25 °C environment to perform reference performance tests. The reference tests include three steps. Firstly, the cells were cycled at a rate of C/25 to obtain dQ/dV curves. Than the standard CC-CV charge and CC discharge at 1C rate was conducted to obtain actual capacity. Lastly the DCR test and the

VR fitting were performed. The reference performance test is performed periodically during cycling test at 75 °C as shown in the inset of Fig. 3a. The flowchart of the test schedule, and the detailed test procedure of dQ/dV test, DCR test and VR test are provided in supporting information (Fig S4).

Three cells were tested: cell 1 and cell 2 were tested with reference performance tests after 0, 50, 100, 200 and 400 cycles at 75 °C. The cell 3 was cycled only at 75 °C without any reference test, the purpose of which was to investigate whether the reference performance test affects the cycle life. Fig. 3a shows the change in capacity retention at 75 °C, and the same fading tendency can be seen on the three cells, indicating that the reference performance test has no significant effect on the cycle life of the battery. Fig. 3b displays the charge-discharge profiles (1C, 25 °C) and the actual capacity of the cycled batteries. In the initial 50 cycles, the capacity loss is very small. But a drastically reduce in capacity and voltage decay is found in the following cycles. From the actual capacity test, it is very hard to get more information about the aging mechanism. The DCR tests provide the information on inner resistance change, which is shown in Fig. 3c. The results show that the resistances experience an uncommon decrease at first 50 cycles and then increase. It is a rare phenomenon in battery aging but we could not get any detailed explanation from DCR. Interestingly, the results

from VR method have the same tendency with DCR test as shown in Fig. 3c. The recorded VR curves are shown in Fig. S5 and the key parameters of R_{total} , V_s , R_o , R_a and R_c are calculated by VR simulation. With the application of VR method, subdivided resistances of R_o , R_a and R_c are obtained and showed in Fig. 3d. Those parameters provide more detailed information related to aging. During the first 100 cycles, both of R_o and R_a decrease. After that, the R_o increases with cycling while the R_a maintains a stable value. While the R_c always increases with cycling. Therefore, it is clear that the reduction of R_{total} during the first 50 cycles is due to the decrease of R_a and R_o . To a large degree, it is affected by the activation process. During which, the electrolyte was filled into the small voids between the particles. Surface components with poor electron conductivity were decomposed and their internal contact and the electron and ion transmission path was strengthened and expanded.

Thus, the charge transfer reaction become easier that contribute to a decline of R_o and R_a during initial 50 cycles. In this stage, the capacity loss was mainly caused by the consumption of recyclable lithium ion (LLI) to form the SEI film. After 50 cycles, the increase in R_o and R_c causes an increase of R_{total} , which is apparently ascribed to the loss of active material (LAM) since that the R_c is mainly affected by the ion diffusion in the active material. Also, the degradation of

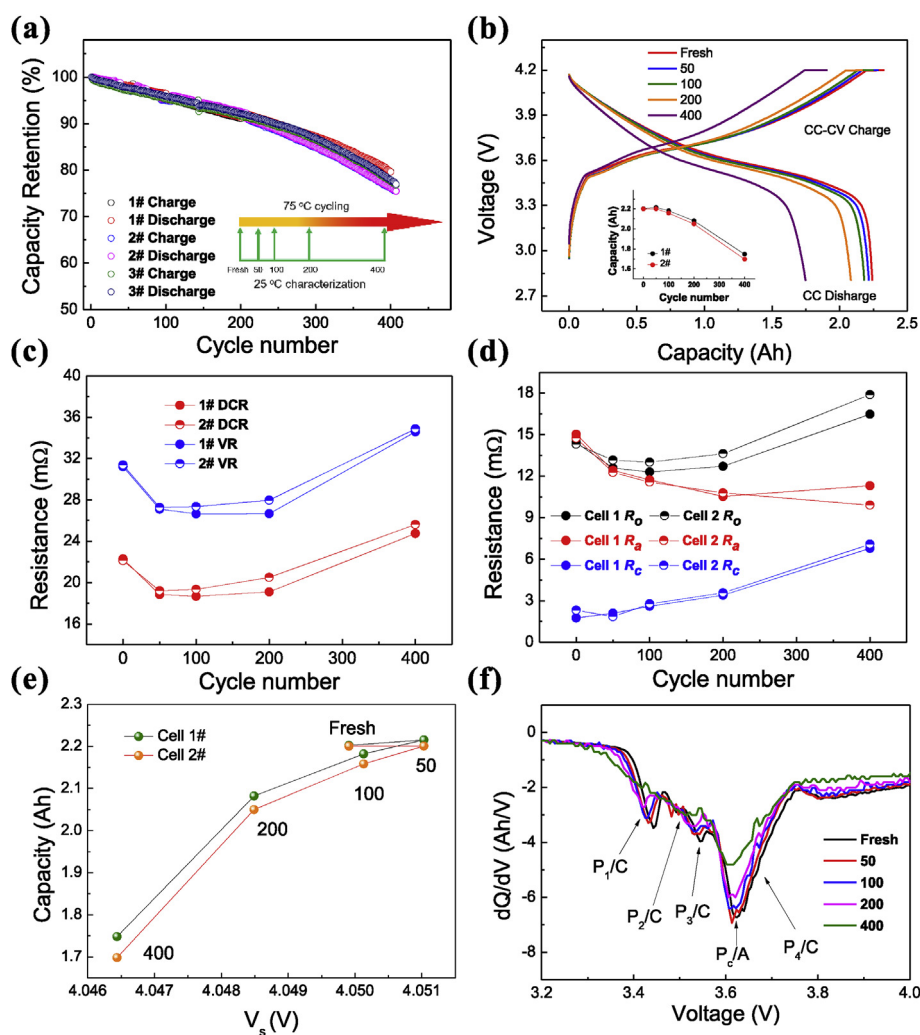


Fig. 3. (a) Capacity retention as function of cycle number at 75 °C. Reference tests including capacity, DCR, incremental capacity and VR characterization were conducted at 25 °C; (b) charge-discharge profile at 1C rate and 25 °C, insert displays the Capacity change with cycling; (c) Comparison of R_{total} by VR simulation and DCR on impedance evaluation; (d) the evolution of parameters R_o , R_a and R_c by VR simulation; (e) dQ/dV curves with different aging states; (f) Actual capacity as function of stable voltage during aging.

active material can lead to the increase of R_0 . In this stage, the capacity loss is mainly caused by the LAM and most of the capacity loss occurs in this stage. The stable voltage (V_s) versus capacity

relationship also display an anomalous behaviour during the initial 50 cycles, which increases with aging. Thus, it is learned from VR method that there are two distinguished capacity loss mechanisms

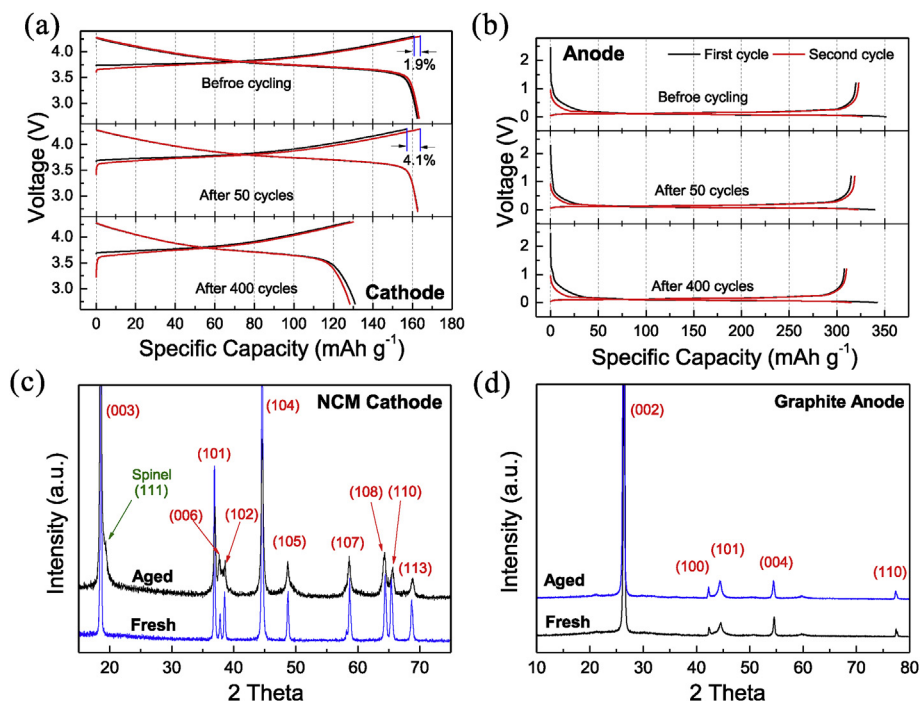


Fig. 4. Half-cells were assembled using electrodes in NCM/graphite full cells before cycling, after 50 cycles and after 400 cycles at 75 °C. The first and second cycles in 1/25C rate of the half-cells: (a) NCM cathode and (b) graphite anode; (c) and (d) XRD results of active materials in fresh and aged electrodes (after 400 cycles).

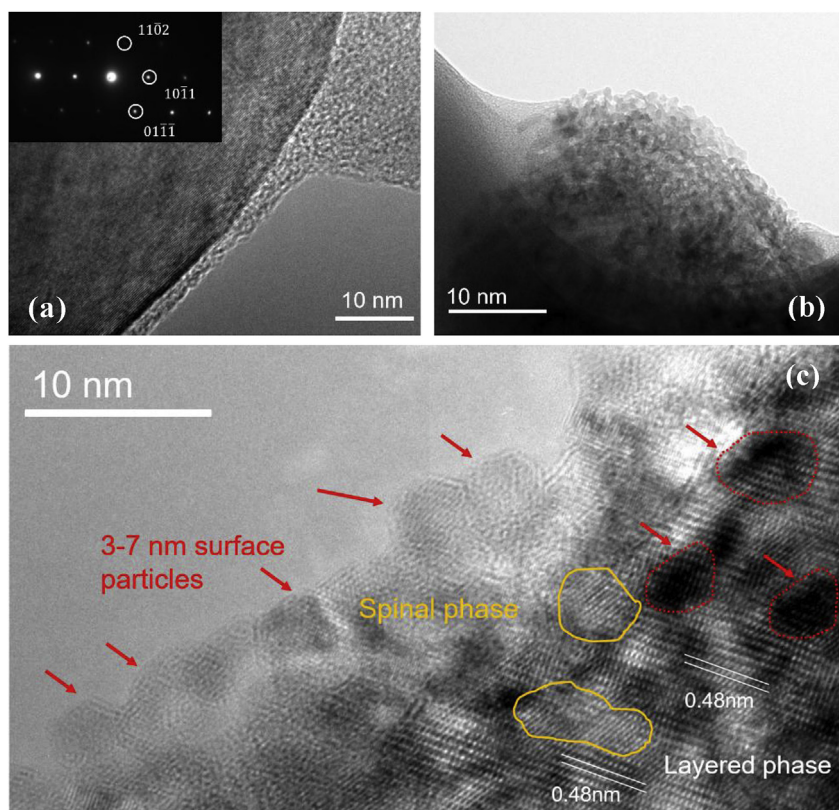


Fig. 5. (a) TEM image of fresh NCM cathode material, inset displays the diffraction pattern of the fresh NCM particle along the $[110]_L$ zone-axis orientation; (b) TEM image of aged NCM cathode material (after 400 cycles at 75 °C); (c) High-resolution TEM image of aged NCM materials in (b).

throughout the whole life of battery. Incremental capacity (dQ/dV) analysis was employed to investigate the aging mechanism. The dQ/dV curves of full cell were plotted against cell voltage over various cycles, as shown in Fig. 3e. To understand the meaning and origin of each dQ/dV peak for the full cell, the incremental capacity of cathode and anode were characterized using half cells. The related discussion can be found in the supporting information in Fig. S7a–d. It can be deduced that the peak of P_c/A at around 3.7 V is due to the lithiation process of NCM cathode. While the peaks of P_1/C , P_2/C , P_3/C and P_4/C derive from the delithiation process of graphite anode. It is noted that during the first 50 cycles, the peak of P_c/A almost remain the same, while the peaks of P_1/C , P_2/C and P_3/C shift apparently, which indicates that capacity loss is caused by the consumption of recyclable lithium ion [28]. As cycling proceeds, the entire dQ/dV peaks shift to lower voltages as well as the peak magnitude decreases with cycles, which suggests that serious LAM in the second stage [31]. The results from dQ/dV are in line with the previous analysis from VR that there are two aging stage during the entire life. Whereas, compared to the long test time of dQ/dV (at least 50 h for one test), VR methods are more time saving and can be applied online.

In order to further confirm the above conclusions, postmodern analysis was carried out. The electrodes in the full cells were disassembled and washed with DMC in an Argon atmosphere, and then reassembled into coin-type half-cells. Fig. 4a and b displays the specific capacities of NCM cathode and graphite anode before, after 50 and 400 cycles. The difference of the second charged capacity to the first charged capacity in cathode indicates the degree of LLI. Clearly, there is more recyclable lithium ion loss (4.1%) after 50 cycles compared to the initial state (1.9%), as shown in Fig. 4a. Nonetheless, both cathode and anode materials recover their capacity when sufficient lithium ions are supplied from the lithium metal anode. Therefore, LLI, not LAM, is responsible for the capacity loss in first stage, just like the VR simulation found. After 400 cycles, the specific capacity of cathode decreased from $\sim 160 \text{ mAh g}^{-1}$ to $\sim 130 \text{ mAh g}^{-1}$, while the anode decreased from $\sim 330 \text{ mAh g}^{-1}$ to $\sim 310 \text{ mAh g}^{-1}$. The specific capacity retention of cathode is 81%, which is much lower than 94% of the anode. The much more serious degradation of cathode suggests that the LAM in cathode is the main reason of capacity loss.

X-ray diffraction (XRD) measurements were conducted with Rigaku D/max 2500/PC using $\text{CuK}\alpha$ radiation with $\lambda = 1.5418 \text{ \AA}$. Fig. 4c and d shows the XRD patterns of the cathode and anode before cycling and after 400 cycles at 75°C . There is no visible change in the shape, strength and position of characteristic diffraction peaks graphite anode, which proves that the crystal structure remain the same after cycling. For the cathode, significant crystallization loss was observed after 400 cycles. The fresh sample possesses strong diffraction peaks which are consistent with the $\alpha\text{-NaFeO}_2$ type structure [32]. The two distinct peak splits observed in the (108)/(110) and (006)/(102) peaks indicate that the NCM material constitutes a well-crystallized layered structure [33]. After cycling, the intensity of all diffraction peaks decreases and the peak width increases, which suggest the degradation of NCM crystal. Even worse, part of the layered structure turned into spinel structure since the new diffraction peak of spinel (111) appears at 19° and the separations of (006)/(102) and (108)/(110) are impaired.

In the TEM observation, the degradation of NCM were characterized at atomic level. As shown in Fig. 5a, the fresh NCM has ordered layered structure, which displays very accurate diffraction spots with $R\bar{3}m$ space group [34]. The particle surface is very clean without any impurities. After 400 cycles, many small particles were formed on the particle surface, which covered the whole NCM crystals (Fig. 5b). In the high-resolution TEM, it can be seen that

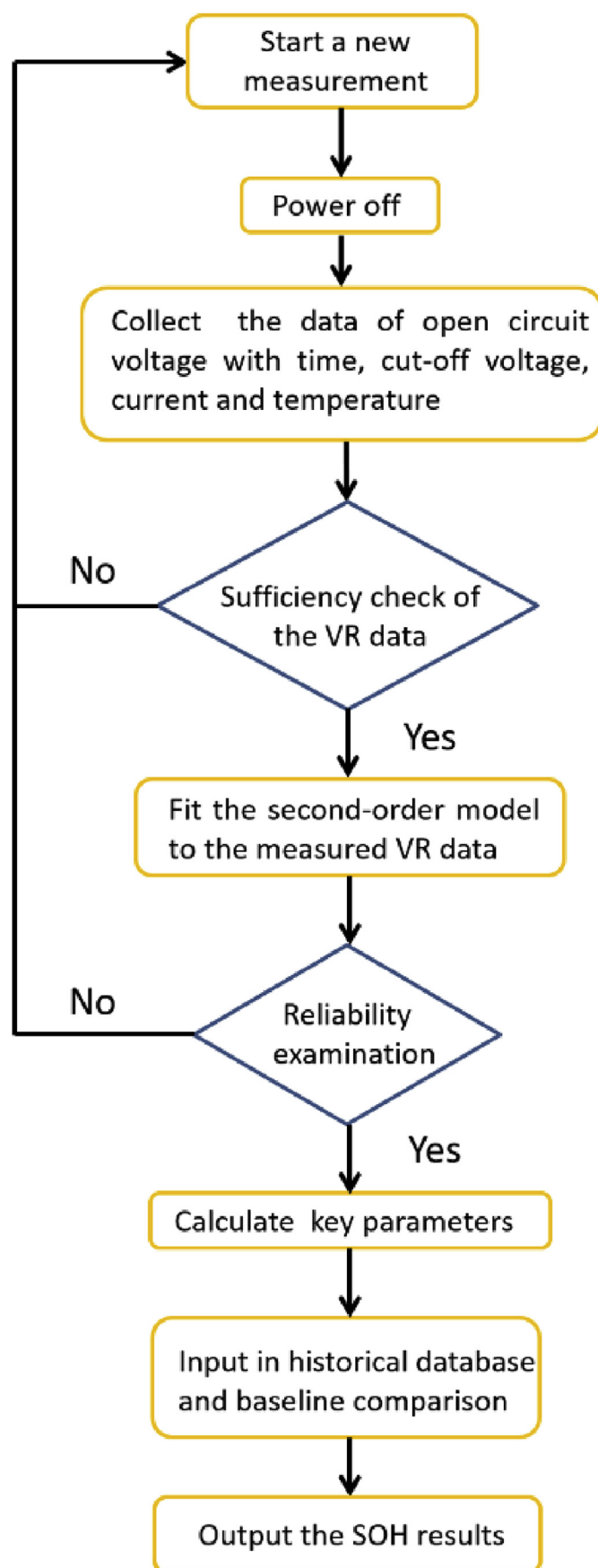


Fig. 6. The overall estimation algorithm by VR simulation.

those particles are 3–7 nm in diameter as shown in Fig. 5c. 7–10 nm amorphous layer coated on the surface and there is spinal phase in the bulk of layered structure. In supporting information (Fig. S8), there are more TEM images on the degradation of layered oxides. This results well agree with the XRD data and previous analysis, which prove the VR simulation is powerful to get more information on aging mechanism analysis.

Herein, we propose the VR strategy for online SOH assessment of the battery. The overall procedure summarized in Fig. 6. For each measurement, new values of the battery VR curve, power-off current and ambient temperature are measured and recorded. The VR data are collected under pre-set state-of-charge and fixed relaxation time. The number of VR data points will be checked before modelling. As soon as the sampling period and measured time were suitable, such as rest time for battery after each charging, the procedure for the SOH evaluation is initiated and the VR simulation is fitted to the actual data. If the precision of curve fitting is acceptable, all the key model parameters (R_{total} , R_o , R_a , R_c , and V_s) are determined and input the database including the previously calculated value. Through baseline comparison and data analysis the SOH can be assessed and output.

5. Conclusions

In summary, the key parameters associated with ohmic depolarization, activation depolarization and concentration depolarization were obtained by VR simulation, which are further used in the SOH evaluation of battery and the identification of battery aging mechanisms. It is verified that the obtained parameters could be correctly correlated with the electrochemical states of the battery and give comprehensive evaluation its SOH. In particular, for the aging mechanism analysis of NCM/graphite pouch cells, two aging stages were easily recognized. The main aging factor in the first stage was ascribed to loss of lithium inventory (LLI), while that in the second stage was due to loss of active material (LAM) in cathode. The proposed VR method and strategy is an excellent candidate for integration into online battery management system applications.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.electacta.2019.02.055>.

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